MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics | **WEEKLY TEST – 02** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Type Questions — MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. If $\sin\theta = x$ and $\sec\theta = y$, then $\tan\theta$ is equal to: [1]

(A) xy (B) $\frac{x}{y}$ (C) $\frac{y}{x}$ (D) $\frac{1}{xy}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (A) xy
Justification: Since $\sec\theta = \frac{1}{\cos\theta}$, we have $\tan\theta = \frac{\sin\theta}{\cos\theta} = \sin\theta \cdot \sec\theta = x \cdot y = xy$.

Q2. In a right-angled triangle ABC, right-angled at C, the value of $\cos(A+B)$ is: [1]

(A) 0 (B) 1
(C) $\frac{1}{2}$ (D) $\frac{\sqrt{3}}{2}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (A) 0
Justification: In $\triangle ABC$, $A+B+C = 180°$. Since $C = 90°$, $A+B = 90°$, so $\cos(A+B) = \cos 90° = 0$.

Q3. If $\sin A = \frac{1}{2}$, then the value of $\cot A$ is: [1]

(A) $\sqrt{3}$ (B) $\frac{1}{\sqrt{3}}$
(C) $\frac{\sqrt{3}}{2}$ (D) 1

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (A) $\sqrt{3}$
Justification: For acute A, $\sin A = \frac{1}{2} \implies A = 30°$, hence $\cot 30° = \sqrt{3}$.

Q4. If $\tan\theta = \dfrac{a}{b}$, then the value of $\dfrac{a\sin\theta - b\cos\theta}{a\sin\theta + b\cos\theta}$ is: [1]

(A) $\dfrac{a^2 - b^2}{a^2 + b^2}$ (B) $\dfrac{a^2 + b^2}{a^2 - b^2}$

(C) $\dfrac{a}{a^2 + b^2}$ (D) $\dfrac{b}{a^2 + b^2}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $\dfrac{a^2 - b^2}{a^2 + b^2}$

Justification: Dividing numerator and denominator by $\cos\theta$: $\dfrac{a\tan\theta - b}{a\tan\theta + b} = \dfrac{a(a/b) - b}{a(a/b) + b} = \dfrac{a^2 - b^2}{a^2 + b^2}$.

Q5. Assertion (A): If $\tan A = \dfrac{4}{3}$, then $\sec A = \dfrac{5}{3}$.

Reason (R): $\sec^2 A = 1 + \tan^2 A$, so $\sec A = \sqrt{1 + (4/3)^2} = \sqrt{25/9} = \dfrac{5}{3}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: For an acute angle A, $\sec A > 0$; thus (R) directly proves (A), and both (A) and (R) are true.

SECTION B: Short Answer Type-I

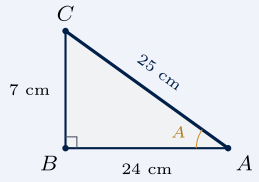
MATHS MANTHAN | $(1 \times 2 = 2$ Marks)

Q6. In $\triangle ABC$, right-angled at B, $AB = 24$ cm and $BC = 7$ cm. Determine the values of: (i) $\sin A, \cos A$, (ii) $\sin C, \cos C$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Find Hypotenuse AC [1 Mark]

In right $\triangle ABC$ ($\angle B = 90°$): $AC = \sqrt{AB^2 + BC^2} = \sqrt{24^2 + 7^2} = \sqrt{576 + 49} = \sqrt{625} = 25$ cm.



Step 2: Ratios for $\angle A$ and $\angle C$ [1 Mark]

(i) $\sin A = \dfrac{BC}{AC} = \dfrac{7}{25}$, $\cos A = \dfrac{AB}{AC} = \dfrac{24}{25}$

(ii) $\sin C = \dfrac{AB}{AC} = \dfrac{24}{25}$, $\cos C = \dfrac{BC}{AC} = \dfrac{7}{25}$

$$\boxed{\text{(i) } \frac{7}{25}, \frac{24}{25} \qquad \text{(ii) } \frac{24}{25}, \frac{7}{25}}$$

Q7. Prove that: $\dfrac{1+\sec A}{\sec A} = \dfrac{\sin^2 A}{1-\cos A}$. **[3]**

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Simplify LHS [1.5 Marks]

$$\text{LHS} = \frac{1+\frac{1}{\cos A}}{\frac{1}{\cos A}} = \left(\frac{\cos A+1}{\cos A}\right)\cdot \cos A = 1+\cos A$$

Step 2: Multiply Numerator & Denominator by $(1-\cos A)$ [1.5 Marks]

$$\text{LHS} = \frac{(1+\cos A)(1-\cos A)}{1-\cos A} = \frac{1-\cos^2 A}{1-\cos A} = \boxed{\frac{\sin^2 \mathbf{A}}{\mathbf{1}-\cos \mathbf{A}} = \text{RHS} \quad \text{[Hence Proved]}}$$

Q8. Prove that: $\sqrt{\dfrac{1+\sin A}{1-\sin A}} = \sec A + \tan A$. **[3]**

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Rationalize Inside the Square Root [1.5 Marks]

$$\text{LHS} = \sqrt{\frac{1+\sin A}{1-\sin A}\cdot\frac{1+\sin A}{1+\sin A}} = \sqrt{\frac{(1+\sin A)^2}{1-\sin^2 A}} = \sqrt{\frac{(1+\sin A)^2}{\cos^2 A}}$$

Step 2: Extract Root for Acute Angle A $(\cos A > 0)$ **& Split** [1.5 Marks]

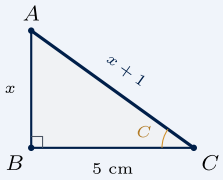
$$\text{LHS} = \frac{1+\sin A}{\cos A} = \frac{1}{\cos A} + \frac{\sin A}{\cos A} = \boxed{\sec \mathbf{A} + \tan \mathbf{A} = \text{RHS} \quad \text{[Hence Proved]}}$$

Q9. In a right $\triangle ABC$, $\angle B = 90°$, $BC = 5$ cm and $AC - AB = 1$ cm. Evaluate $\dfrac{1+\sin C}{1+\cos C}$. **[3]**

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Determine Sides of Right $\triangle ABC$ [1.5 Marks]

Let $AB = x$ cm, so $AC = (x+1)$ cm. By Pythagoras' theorem:



$$(x+1)^2 = x^2 + 5^2 \implies x^2 + 2x + 1 = x^2 + 25$$

$$2x = 24 \implies x = 12 \text{ cm}$$

Hence $AB = 12$ cm and $AC = 13$ cm.

Step 2: Evaluate Required Expression [1.5 Marks]

For angle C: $\sin C = \dfrac{AB}{AC} = \dfrac{12}{13}$ and $\cos C = \dfrac{BC}{AC} = \dfrac{5}{13}$.

$$\frac{1+\sin C}{1+\cos C} = \frac{1+\frac{12}{13}}{1+\frac{5}{13}} = \frac{25/13}{18/13} = \boxed{\mathbf{\frac{25}{18}}}$$

Q10. ADA Compliant Wheelchair Access Ramp Architecture

According to architectural accessibility standards, an entrance ramp for wheelchairs must have a gentle incline. An architect designs a straight access ramp PR to bridge a vertical doorway threshold rise of $RQ = 0.6$ m above the ground level. The ramp is inclined at an angle α to the horizontal ground such that $\tan\alpha = \dfrac{1}{12}$. **Based on the above geometric model, answer the following questions:**

(i) Determine the horizontal run PQ of the access ramp along the ground. [1]

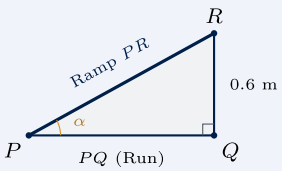
(ii) Calculate the exact slant length PR of the ramp surface (leave in surd form or round to two decimal places). [1]

(iii) If the ramp slope is redesigned to an angle of 30° for a secondary emergency ramp of the same vertical rise 0.6 m, find the new required ramp surface length. [2]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Sub-part (i): Horizontal Run PQ [1 Mark]

$$\tan\alpha = \frac{RQ}{PQ} \implies \frac{1}{12} = \frac{0.6}{PQ} \implies \boxed{PQ = \mathbf{7.2}\text{ m}}$$



Sub-part (ii): Exact Slant Length PR [1 Mark]

$$PR = \sqrt{PQ^2 + RQ^2} = \sqrt{7.2^2 + 0.6^2} = 0.6\sqrt{12^2 + 1^2}$$

$$= \mathbf{0.6}\sqrt{\mathbf{145}}\text{ m} \approx \boxed{\mathbf{7.22}\text{ m}}$$

Sub-part (iii): Redesigned Emergency Ramp at 30° [2 Marks]

For angle 30° and vertical rise 0.6 m, let the new surface length be PR':

$$\sin 30^\circ = \frac{0.6}{PR'} \implies \frac{1}{2} = \frac{0.6}{PR'} \implies \boxed{PR' = \mathbf{1.2}\text{ m}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★